Auteur: Ahmed Meiyaki
Studierichting: Informatica
Oefeningenreeks 2E nr. 18.6

$$
\begin{aligned}
&4 \cdot \log_x \sqrt{3} + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} + \frac{1}{3} = 0 \\
&\Leftrightarrow 2 \cdot \log_x 3 + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} = -\frac{1}{3} \\
&\Leftrightarrow 2 \cdot \log_x 3 + \log_x 3^{\frac{1}{3}} + \log_x \frac{1}{9} = -\frac{1}{3} \\
&\Leftrightarrow \log_x \frac{9}{9} + \log_x 3^{\frac{1}{3}} = -\frac{1}{3} \\
&\Leftrightarrow \log_x 3 = -1 \\
&\Leftrightarrow x = \frac{1}{3}
\end{aligned}
$$

References